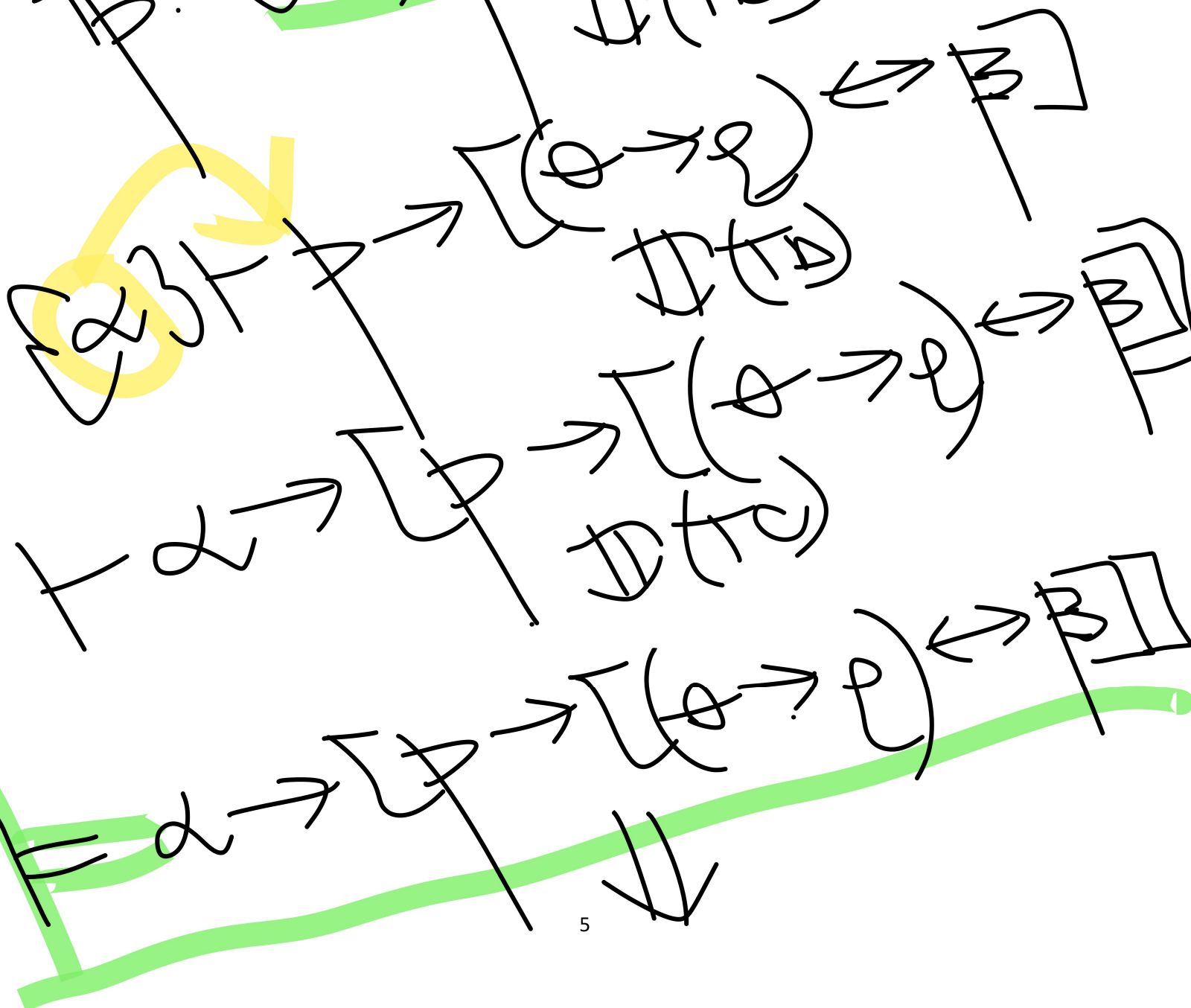


$\Gamma \vdash \alpha \Rightarrow \beta$

The Lr: $\Gamma \vdash \alpha \Rightarrow \beta \Rightarrow \Gamma \vdash \alpha \Rightarrow \beta$



$$\begin{aligned}
 & \text{Int} \models \alpha \rightarrow \neg \beta \rightarrow \neg (\beta \rightarrow \phi) \Leftrightarrow \neg \beta \\
 & \text{Int}(\alpha \rightarrow \neg \beta \rightarrow \neg (\beta \rightarrow \phi)) \Leftrightarrow \neg \beta \\
 & \quad \quad \quad \parallel \quad \quad \quad = \neg (*)
 \end{aligned}$$

$$\begin{aligned}
 & \text{Int}(\alpha) \rightarrow \neg \text{Int}(\beta) \rightarrow \neg \text{Int}(\beta \rightarrow \phi) = \text{Int}(\neg \beta) \\
 & \rightarrow \neg \text{Int}(\neg \beta) \rightarrow \neg \text{Int}(\beta) \Leftrightarrow \neg \text{Int}(\beta)
 \end{aligned}$$

$$\begin{aligned}
 & \text{Int}(\alpha \wedge \neg \beta) \rightarrow (\neg \beta \rightarrow \phi) \\
 & = (\text{Int}(\alpha) \wedge \neg \text{Int}(\beta)) \rightarrow (\neg \beta \rightarrow \phi) \\
 & \quad \quad \quad \text{Int}(\alpha) \rightarrow \neg \text{Int}(\beta) \rightarrow \neg \beta \rightarrow \phi
 \end{aligned}$$

$$\begin{aligned}
 & (*) \Rightarrow \text{Int}(\alpha) \rightarrow \neg \text{Int}(\beta) \rightarrow \neg \text{Int}(\beta \rightarrow \phi) \\
 & \rightarrow \neg \text{Int}(\beta) \Leftrightarrow \neg \text{Int}(\beta)
 \end{aligned}$$

Case 1: $\alpha \in \mathcal{L}$. $\neg(\alpha \wedge \neg \alpha)$

$$\Rightarrow \neg(\alpha \wedge \neg \alpha) \rightarrow (\neg \alpha \rightarrow \alpha) \quad \text{tautology}$$

$$\equiv 0 \rightarrow (\neg \alpha \rightarrow \neg \alpha) = 1$$

$$\equiv 0 \rightarrow (\neg \alpha \rightarrow \neg \alpha) = 1$$

$$\equiv 1$$

Case 2: $\alpha \in \mathcal{L}$. $\neg(\alpha \wedge \neg \alpha)$

$$\Rightarrow \neg(\alpha \wedge \neg \alpha) \rightarrow (\neg \alpha \rightarrow \alpha) \quad \text{tautology}$$

$$\Rightarrow \neg(\alpha \wedge \neg \alpha) \rightarrow (\neg \alpha \rightarrow \alpha) = 1$$

func:

$$\begin{aligned}
 & \left(\begin{array}{c} * \\ * \\ * \end{array} \right) \Leftrightarrow \tilde{\ln}(\alpha) \leq \ln(\beta) \rightarrow \\
 & \rightarrow (\ln \varphi) \leftrightarrow \ln(\beta) \Leftrightarrow \ln(\beta) = 1
 \end{aligned}$$

$$\Leftrightarrow \ln(\beta) \rightarrow (\ln \varphi \leftrightarrow 1) = 1$$

$$\Leftrightarrow \ln(\beta) \leq \ln(\varphi) \leftrightarrow 1 = 1$$

$$= (\ln \varphi) \rightarrow 1 \wedge$$

$$\wedge (1 \rightarrow \ln \varphi) =$$

$$= (\ln \varphi \vee 1) \wedge (1 \vee \ln \varphi) = 1$$

$$= 1 \wedge \ln(\varphi) = \ln(\varphi) \Rightarrow$$

$$(\neg A), (\neg B) \Rightarrow (\neg(A \wedge B) \rightarrow \neg A)$$

$$(\neg(A \wedge B) \rightarrow \neg A) \Rightarrow (\neg(A \wedge B) \rightarrow \neg B)$$

$$(\neg(A \wedge B) \rightarrow \neg A) \Rightarrow (\neg(A \wedge B) \rightarrow \neg B)$$

$$\Rightarrow$$

$$\neg(A \wedge B)$$

$$\neg(A \wedge B)$$

$$\neg(A \wedge B) \rightarrow \neg A$$

$$\neg(A \wedge B) \rightarrow \neg A$$

$$\neg(A \wedge B)$$

$$\neg(A \wedge B)$$

$$\neg(A \wedge B)$$

$$\neg(A \wedge B)$$

$$\neg(A \wedge B)$$

$$\neg(A \wedge B)$$

$$\neg(A \wedge B) \rightarrow \neg A$$

